KFUPM · ICS 483

WaveFlow-UIE

A Single-Pipeline **Frequency-Guided Flow Model** for Underwater Image Restoration

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April 2026

Agenda

What we'll cover today

01
The Problem
Underwater degradation & why it matters

02
Background
Frequency-domain insights & WF-Diff baseline

03
Our Method
WaveFlow-UIE — a single wavelet-flow pipeline

04
Experiments
Quantitative, qualitative & speed analysis

Underwater Images Are Degraded

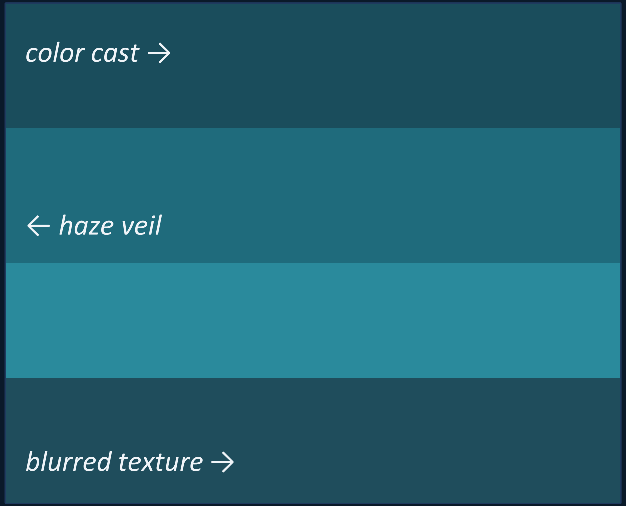
And that breaks downstream computer-vision systems

Light absorption & scattering

produce **color cast**, **low contrast**, **haze**, and **lost texture detail**.

WHY IT MATTERS

- Marine inspection & biology
- Underwater robotics & navigation
- Object detection & tracking



Degraded underwater scene

Baseline: WF-Diff (CVPR 2024)

Strong frequency-aware results — but a heavy two-stage pipeline

STAGE 1

WFI²-net

Coarse wavelet-domain enhancement via Transformer + Fourier blocks

→

STAGE 2

FRDAM (diffusion)

Two diffusion branches refine LL & HF residuals separately

THE COST

Two networks + iterative diffusion sampling → ~0.28s / image inference and high architectural complexity.

Our Idea: Collapse the Pipeline

Replace coarse-then-refine with a single learned flow

WF-Diff

two-stage

Coarse Net (WFI²-net)

Diffusion Branch (LL)

Diffusion Branch (HF)

VS

WaveFlow-UIE

single pipeline

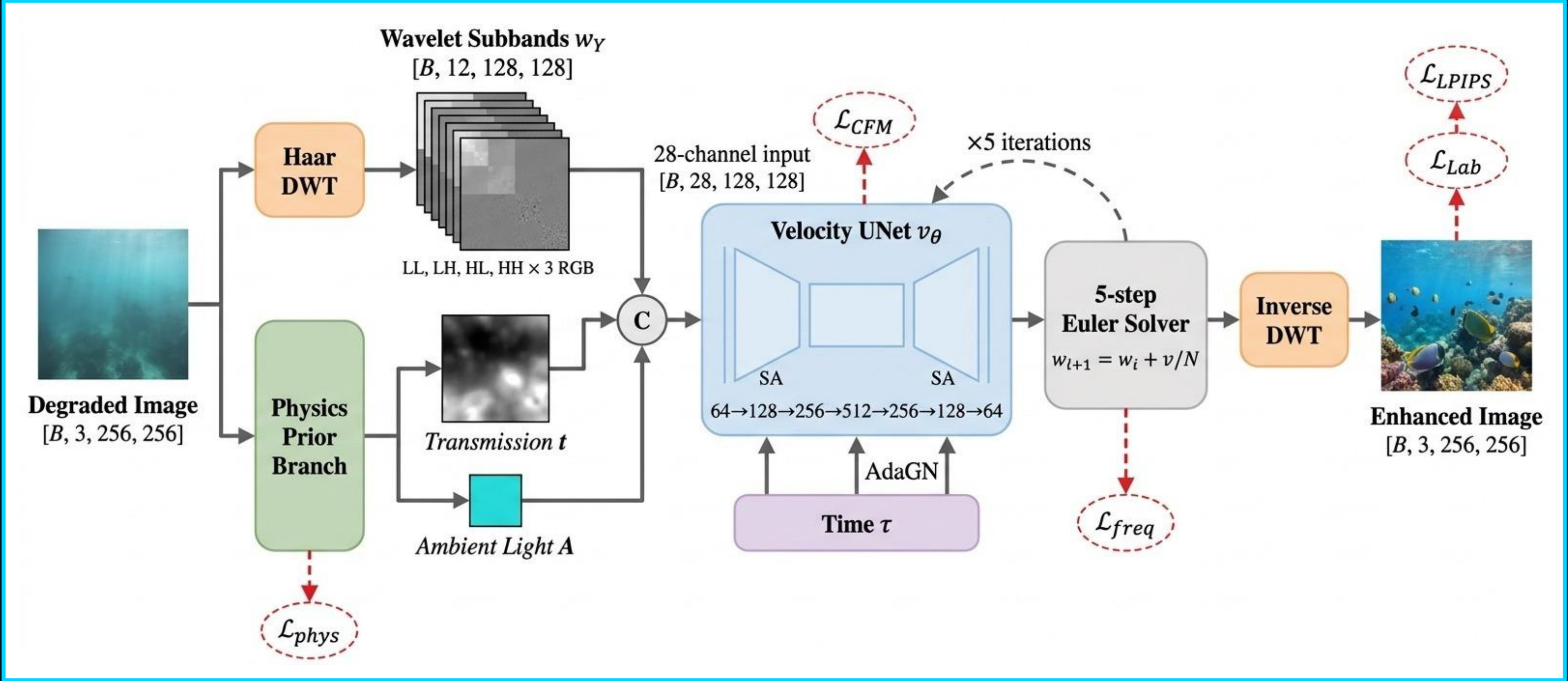
Conditional Velocity Network

learns wavelet-space flow direction

One network. Frequency-aware. Physics-conditioned. Trained with flow matching.

Detailed Architecture

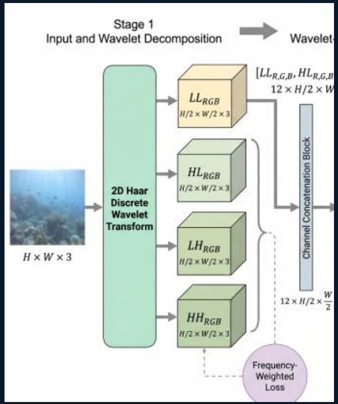
Tensor shapes, sub-band layout, and loss attachment points



Inputs (image) → **Wavelet + Physics** → **Velocity UNet × 5** → **IDWT** · *red dashed = loss supervision*

Inside the Wavelet Decomposition

Splitting the image into bands the model can reason about separately



Input $x \in \mathbb{R}^{3xHxW}$

DWT

Haar wavelet

LL *color*	**LH** *horiz*
HL *vert*	**HH** *diag*

4 sub-bands · H/2 × W/2

W_x

12 channels

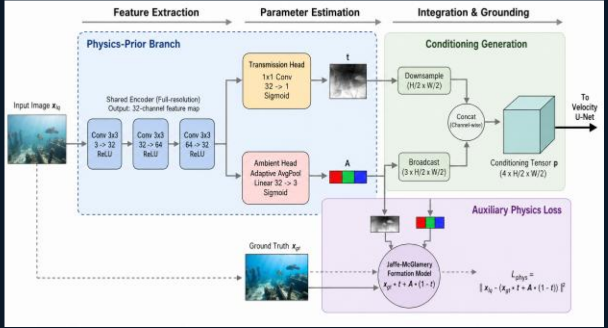
(LL + 3·HF) × 3 RGB

→ fed to velocity net

Why this helps: degradation is **non-uniform across bands** · color sits in LL, fine detail in HF — process them with different weights.

Inside the Physics Prior

4 conv layers + 2 heads — predicts transmission t and ambient light A



Input x

3 ch · H × W

→

Conv 3×3	Conv 3×3	Conv 3×3
3 → 32 ch + ReLU	32 → 64 ch + ReLU	64 → 32 ch + ReLU

BACKBONE · 3 conv layers

→ **Conv 1×1**
32 → 1 + Sigmoid

t · transmission map
(B, 1, H, W) · per-pixel

→ **AvgPool → FC**
32 → 3 + Sigmoid

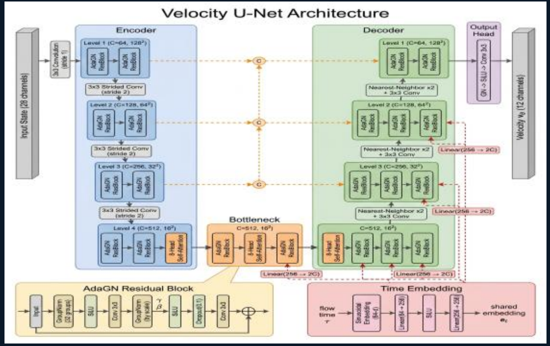
A · ambient light
(B, 3) → broadcast

p = [t, A] → **4 channels**, downsampled to **H/2 × W/2** · fed to velocity UNet

Underwater formation model: $x \approx y \cdot t + A \cdot (1 - t)$

Inside the Velocity Network

AdaGN U-Net · 64 → 128 → 256 → 512 · self-attention at 16²



U-NET

4 levels · 3 down / 3 up

Resolution	Encoder		Decoder
128²	64	- - -	64
64²	128	- - -	128
32²	256	- - -	256
16²	512	- - -	512 ◆ SA

ENCODER — DECODER

AdaGN RESBLOCK

how time τ enters every block

- GroupNorm (32 groups)
- SiLU
- Conv 3×3
- **AdaGN · scale & shift ← τ**
- SiLU + Dropout 0.1
- Conv 3×3

τ → *sinusoidal(64)* → *MLP* → *256-d*

◆ self-attention at 16² · 8 heads × 64 dim · **skip = concat** · down: stride-2 conv · up: nearest + conv

INPUT **28 ch** (12 w_t + 12 w_x + 4 [t,A]) · **OUTPUT** **12 ch** velocity $\hat{v}$ ·

Training Objective

Five complementary losses guide what the flow learns

$$\mathcal{L}total = \lambda_1 Lcfm + \lambda_2 Lfreq + \lambda_3 Llpips + \lambda_4 Llab + \lambda_5 Lphys$$

ℒ_cfm

Flow Matching

predict the right velocity in wavelet space

ℒ_freq

Frequency

weighted L1 — extra weight on high-freq detail

ℒ_lpips

Perceptual

deep-feature similarity (LPIPS)

ℒ_lab

Color (Lab)

consistent color in perceptual color space

ℒ_phys

Physics

$x \approx y \cdot t + A \cdot (1 - t)$

Frequency loss carries the largest weight on high-frequency components — preserves edges & texture

Datasets & Setup

Trained and evaluated across four standard underwater benchmarks

UIEB

890

images

real-world · paired

800 train · 90 test

LSUI

4279

images

diverse scenes · paired

3879 train · 400 test

UFO-120

120

images

USR-248-style · paired

test set

EUVP

515

images

Images · paired

test set

EVALUATION SETUP

RESOLUTION

256×256 & native

METRICS

FID · LPIPS · PSNR · SSIM

NON-REFERENCE

UIQM · UCIQE on U45 / C60

HARDWARE

single GPU · consistent runtime

Hyperparameters & Setup

Reproducibility — what we trained, how, and where

Training

Optimizer	**Adam (β_1=0.9, β_2=0.999)**
Learning rate	**1×10^{-4}**
Batch size	**8**
Resolution	**256 × 256**
Epochs	**200**
Scheduler	**Cosine decay**

Loss Weights

λ_cfm	1.0	*flow matching*
λ_freq	0.5	*frequency L1*
λ_lpips	0.1	*perceptual*
λ_lab	0.2	*Lab color*
λ_phys	0.05	*physics consistency*
α (HF)	2.0	*high-freq weight*

Inference + HW

Flow steps K	**5**
Step size Δt	**1 / K**
Inference time	**~79 ms (256^2)**
GPU	**RTX 5080 (16GB)**
Framework	**PyTorch 2.x**
Wavelet	**Haar**

Quantitative Results

Best-or-second-best across UIEB, LSUI, UFO-120 and EUVP

Table 2: Native-resolution quantitative comparison on UIEB and LSUI.

Method	UIEB						LSUI					
	PSNR↑	SSIM↑	LPIPS↓	FID↓	UCIQE↑	UIQM↑	PSNR↑	SSIM↑	LPIPS↓	FID↓	UCIQE↑	UIQM↑
UIEWD	14.3053	0.7507	0.3945	97.1222	0.5622	3.8048	17.4945	0.7853	0.3582	55.0274	0.5600	**4.2795**
UWCNN	19.1571	0.8601	0.3312	74.7177	0.5378	3.3864	20.6892	0.8354	0.3660	54.3170	0.5530	3.7243
WaterNet	23.1406	0.9156	0.1833	36.1890	0.5806	**3.8336**	24.4207	0.8768	0.2558	38.1637	**0.5856**	4.2284
WF-Diff	23.6565	0.9346	0.1963	38.5893	**0.5831**	3.8225	25.9974	**0.8973**	0.1957	27.5106	0.5699	4.0466
Ours	**24.6786**	**0.9350**	**0.1589**	**27.6999**	0.5820	3.6330	**27.2355**	0.8949	**0.1519**	**27.0609**	0.5728	4.1128

Ablation Study

What does each component actually contribute?

Configuration	PSNR ↑	SSIM↑	LPIPS↓	FID↓	UCIQUE↑	UIQM↑
Full model (Ours)	**24.9818**	**0.9316**	**0.1390**	**29.7382**	0.5835	4.0450
No physics prior branch	24.5000	0.9234	0.1469	35.7000	**0.5853**	4.0912
No frequency loss weighting	24.0902	0.9298	0.1426	33.8364	0.5852	4.0270
No LPIPS loss	24.0288	0.9047	0.1574	35.1146	0.5753	**4.2561**
No lab color loss	24.1482	0.9220	0.1848	44.0632	0.5819	4.0640
K = 1 (single step)	24.9382	0.9306	0.1403	30.6582	0.5827	4.0638

WHAT WE EXPECT TO SEE

Removing the physics prior hurts color accuracy. Removing the frequency loss softens fine detail. K=1 trades quality for speed.

Fill in the dashes once final ablation runs are complete.

Qualitative Results

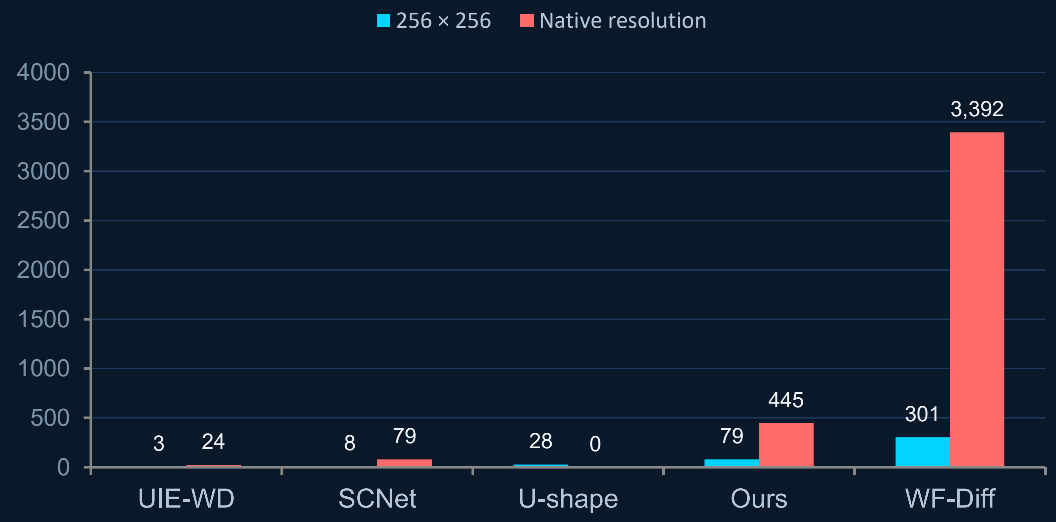
Restored color and detail across diverse underwater scenes

Input | WF-Diff | WaveFlow-UIE

Speed & Tradeoffs

Faster than diffusion-based WF-Diff — honest about where we sit

Inference time · ms / image ↓ (UIEB, log scale)



3.8× / 7.6× faster

at 256^2 / native res · vs WF-Diff

- Single-pipeline simplicity
- Best FID on UIEB / UFO-120
- Best LPIPS & SSIM on LSUI
- Slower than purely-feedforward CNNs (still iterative)
- Native-resolution cost grows with image size

Takeaways & Future Work

What we showed — and where it goes next

CONTRIBUTIONS

- Reformulated WF-Diff as a single wavelet-domain flow
- Added a physics prior branch for transmission & ambient
- Frequency-aware losses: detail + color jointly preserved
- Faster inference at competitive — often better — quality

FUTURE WORK

- Reduce flow steps (1-step distillation)
- Test on real-time underwater video streams
- Evaluate on downstream detection / SLAM
- Stronger physics priors (depth-aware)

Thank you.

Questions?